Calculation

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In the previous section, you have derived a mathematical model for $Z(t)$, the mass of the pollutant in the lake:

$$\frac{dZ}{dt} = -\frac{Q}{V_{lake}} Z(t) + C_{in} Q + U, \quad Z(0) = C_{in} V_{lake},$$

where Q, V_{lake}, C_{in} and U all have constant values.

In this section, you will not solve the differential equation analytically (which is not that difficult, but it is beyond the scope of this course). You will also not approximate the solution numerically (as you will learn to do in the next section). In this section, you are going to investigate the solution just by using the phase line.

Exercise A

1/1 point (graded)

The equilibrium value Z_e for Z is

- $Z_e = V_{lake}(C_{in} + \frac{Q}{U})$
- $Z_e = V_{lake}\frac{U}{Q}$
- $Z_e = V_{lake}(\frac{U}{Q} - C_{in})$
- $Z_e = V_{lake}(C_{in} + \frac{U}{Q})$ ✔

Answer

Correct: Do you recognise the expression for Z_e when you set $U = 0$?

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✔ Correct (1/1 point)

Exercise B

1/1 point (graded)

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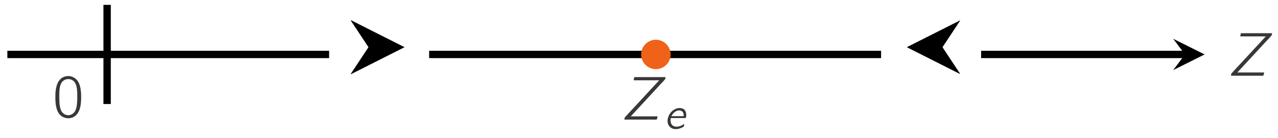
The differential equation for the mixing problem is autonomous. So, a phase line can be drawn. First, draw graph of the right hand side of the differential equation as a function of Z. Then, complete the phase line below, by adding the correct arrow heads to the correct locations.



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FEEDBACK

✔ Correctly placed 2 items.

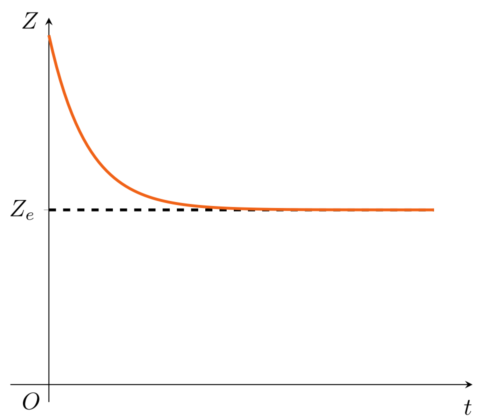
✔ Excellent, you have correctly created the phase line.

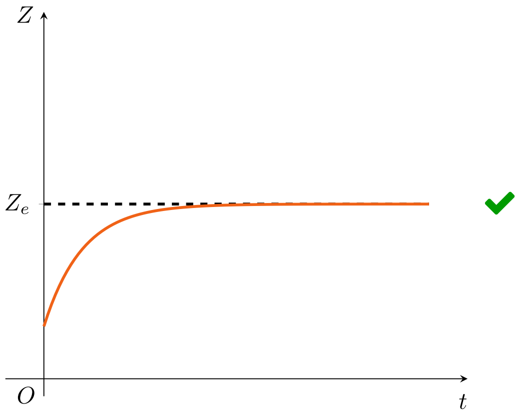
✔ Your highest score is 1.0

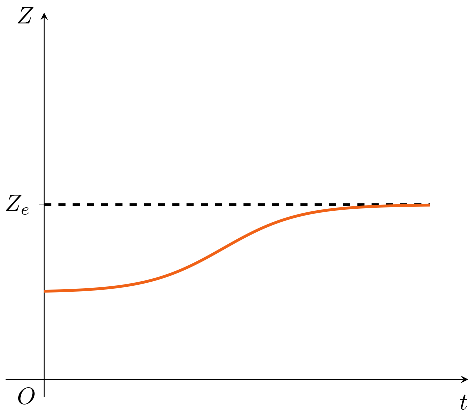
Exercise C

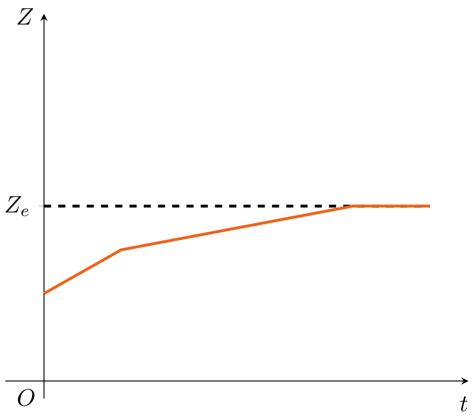
1/1 point (graded)

Which of the graphs below could depict the (exact) solution $Z(t)$?









Answer

Correct:

Indeed: $Z(0)$ is smaller than Z_e, $Z(t)$ increases for all time t, and the derivative is highest for the smallest Z and decreases later.

Submit You have used 2 of 2 attempts

Answers are displayed within the problem

In the next section, you will see whether the question about the pollution by the factory has been answered.